Clinical Decision Support System for Disease Risk Assessment

Progress Report - 3

Submitted By: Akiful Mohaimin (300369823)

Course: CSIS 4495 – Applied Research Project

Section: 03

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1. Work Date / Hours Log

Date	Hours	Description of Work Done
Mar 3, 2026	3.0	Prepared the Cleveland Heart Disease Dataset and verified clinical feature columns for machine learning modelling.
Mar 5, 2026	3.5	Implemented baseline Logistic Regression and Random Forest models for the Cleveland dataset using cleveland_model.py.
Mar 7, 2026	4.0	Generated model evaluation outputs including confusion matrices (cleveland_lr_confusion_matrix.png,cleveland_rf_confusion_matrix.png) and ROC visualizations.
Mar 8, 2026	4.0	Collaborated with Uthara on final model optimization including cross-validation, hyperparameter tuning and feature importance analysis for the Random Forest model.

2. Summary Description of Work Done During This Reporting Period

This reporting period was dedicated to the work on the implementation and evaluation of heart disease prediction models using Cleveland Heart Disease data. The data was checked and made ready to machine learning modelling such that all the columns of clinical features were properly formatted and ready to feed into the model.

Two classification models were used and tested with the help of classification metrics such as accuracy, precision, recall, and F1-score: the Logistic Regression and the Random Forest ones. The predictive power of the Cleveland dataset was studied with the help of these models and the identification of the key clinical predictors of heart disease.

Further results of the evaluation such as the confusion matrices and ROC curve illustrations were produced to gain a better insight into the model performance and classification behaviour. Such visualizations were able to compare the efficacy of models applied.

Using cross-validation and hyperparameter tuning together with Uthara, the Random Forest model was optimized to achieve better predictive results. The only slight problem that was encountered at this step was the synchronization of features of the dataset with the format needed by the required model that was addressed by further preprocessing and validation of the features. The obtained results will justify the next stage of the development implementation of the trained model into the clinical decision support system.

3. Repo Check-In of Implementation Completed

During this reporting period, repository updates were committed related to the Cleveland heart disease dataset modelling and evaluation process. These updates include model implementation scripts and visualization outputs generated during model evaluation.

The following files were added or updated in the repository:

Model Training Script

- cleveland_model.py

Dataset

- heart_cleveland_upload.csv

Evaluation Outputs

- cleveland_lr_confusion_matrix.png
- cleveland_rf_confusion_matrix.png
- cleveland_roc_curve_comparison.png

Feature Importance Outputs

- cleveland_rf_feature_importance.csv
- cleveland_rf_feature_importance.png

These outputs provide evaluation results and feature importance analysis for the Cleveland heart disease dataset, allowing comparison with the Heart Failure dataset modelling results.

4. AI Use Section

AI Tool	Version	Purpose of Use	Value Added by User
ChatGPT	GPT-5 (Premium)	Understanding feature importance in Random Forest models	Implemented feature importance extraction and visualization to identify the most influential clinical indicators contributing to heart disease prediction.
ChatGPT	GPT-5 (Premium)	Guidance on generating evaluation visualizations such as ROC curves and confusion matrices	Applied the suggested methods to generate ROC curve and confusion matrix visualizations to evaluate and compare classification model performance.

5. APPENDIX

5.1 Appendix A – AI Prompts

1. How can I determine and visualize feature importance in a Random Forest model using Python
2. How can I generate ROC curve and confusion matrix visualizations to evaluate classification model performance in Python

5.2 Appendix B – Feature Importance

Feature importance analysis was conducted using the Random Forest model trained on the Cleveland Heart Disease dataset. Important predictors included **age, cholesterol, maximum heart rate, and resting blood pressure**.

The results were exported as both a table (`cleveland_rf_feature_importance.csv`) and visualization (`cleveland_rf_feature_importance.png`). These outputs help highlight the clinical variables that most strongly influence heart disease prediction.

5.3 Appendix C – Cleveland Model Results

```
==================================================
LOGISTIC REGRESSION
==================================================
Accuracy: 0.9166666666666666
              precision    recall  f1-score   support

           0       0.86      1.00      0.93        32
           1       1.00      0.82      0.90        28

    accuracy                           0.92        60
   macro avg       0.93      0.91      0.91        60
weighted avg       0.93      0.92      0.92        60


==================================================
RANDOM FOREST
==================================================
Accuracy: 0.8666666666666667
              precision    recall  f1-score   support

           0       0.82      0.97      0.89        32
           1       0.95      0.75      0.84        28

    accuracy                           0.87        60
   macro avg       0.89      0.86      0.86        60
weighted avg       0.88      0.87      0.86        60
```